

December 9, 2025
California Coastal Commission
45 Fremont Street, Suite 2000
San Francisco, CA 94105

Re: Public Comment on Application No. 9-25-0512 – City of Fort Bragg Desalination Buoy Pilot Project (Hearing Date: December 11, 2025)

Dear Commissioners,

My name is Sarayu Ramnath. I am a lifelong California resident and a graduate student focused on ecosystem-based adaptation solutions. I have both personal and professional experience with our coastal ecosystems, and understand the potential role of innovative water technologies like the Iceberg buoy in advancing sustainable water management in California. However, even small-scale pilot projects must be implemented with rigorous safeguards to ensure the protection of communities that rely on these ecosystems.

I am submitting comments in cautious support of the City of Fort Bragg's application for a 12-month pilot deployment of the Oneka Desalination Buoy (Iceberg) wave-powered desalination buoy. My support is contingent upon two conditions:

1. The inclusion of structured, environmental monitoring before and during deployment.
2. The establishment of a hard-substrate mitigation fund should the buoy or its pipelines disturb sensitive benthic habitat.

Project Context

The Iceberg is a singular wave-powered device that converts seawater to freshwater 0.5 miles offshore in Mill Bay, and delivers it via submerged pipeline to the City of Fort Bragg's (hereafter referred to as the City) wastewater treatment plant, all without using electricity from land based sources⁶. This single-buoy pilot is expected to produce an average of 13,200 gallons per day over a 12-month period⁴. Preferred deployment areas have been noted as soft-bottom habitat to minimize impacts on marine life, and the pipeline alignment is designed to follow previously disturbed habitat near the wastewater treatment plant's ocean outfall⁶. This single-buoy pilot will collect quantitative data that could inform the design and permitting of future utility-scale deployments to support the City's water portfolio⁴.

1. Requirement for Comprehensive Monitoring

Although modest in scale, this pilot must include structured, independent, third-party environmental monitoring before, during, and after deployment. Monitoring should track potential impacts to soft-bottom benthic communities, trapping risk for plankton, area usage by foraging marine wildlife, and any unexpected disturbances associated with mooring or pipeline movements. Pre-deployment surveys are essential to establish a baseline to assess project-related effects. Future larger-scale buoy arrays would yield a much larger environmental footprint, which is why empirical data from the pilot should directly inform CEQA review, including if a full Environmental Impact Report (EIR) is needed.

This oversight is especially important because the project is proceeding under an Initial Study/Mitigated Negative Declaration (IS/MND) ⁸, which is a CEQA pathway that evaluates likely impacts and identifies avoidance and minimization measures but does not provide the depth of analysis, alternatives evaluation, or cumulative impact assessment required of a full Environmental Impact Report ⁵. Given that the pilot will generate the first empirical data on biological, oceanographic, and operational effects of this technology in the state ⁷, the monitoring program should be comprehensive enough to inform future permitting, including whether a utility-scale array would require preparation of a full EIR.

The IS/MND determined that all potential impacts could be reduced to less-than-significant levels with mitigation ⁸, so the pilot would function as a test of those assumptions. Although the IS/MND evaluates broad categories such as biological resources, hydrology, water quality, and tribal cultural resources ⁸, there is no evaluation of how these offshore buoys would interact with migrating wildlife, soft-bottom organisms, or local oceanography. These gaps highlight the need for robust environmental review. A publicly transparent monitoring plan, including baseline surveys, would ensure that findings from this pilot meaningfully support long-term policy decisions, rather than rely on modeled or anticipated outcomes.

2. Necessity for Hard-Substrate Mitigation Fund

I understand that the California Coastal Commission (CCC) has previously required hard-substrate mitigation fees in similar marine development projects ³. Although the City plans to site the buoy in soft-substrate habitat to minimize impacts, the environmental setting described in the IS/MND describes that the project area is not exclusively soft-substrate ⁸. Soft-sediment areas in Fort Bragg's nearshore waters are closely mixed with rocky reef and kelp forest habitats, supporting some of the region's highest biodiversity and are critical to the recovery of North Coast kelp ecosystems ⁹. Recent surveys in the IS/MND also document a

long-term shift from sand to exposed rocky substrate and the presence of sensitive species such as red abalone, sea urchins, and foliose algae⁸. Even projects designed for soft-bottom placement can unintentionally impact hard-substrate communities during cable anchoring, pipeline installation, or vessel operations¹⁰. This is why I urge the Commission to require the City of Fort Bragg to provide funds for hard-substrate mitigation if sensitive benthic habitat is disturbed.

For precedent, the CCC has applied this mechanism in prior projects, including Southern California Edison's mitigation for the San Onofre Nuclear Generating Station, which funded the Wheeler North reef, an artificial reef and oversight by CCC scientists to compensate for impacts to the marine environment¹. Establishing this fund would also allow for adaptive management, supporting ongoing monitoring and mitigation efforts². If no such fund is yet defined for this project, I request that the Commission commit to establishing one, benefitting benthic habitat recovery in the area. Although this is a single-buoy pilot, setting up a mitigation fund now will provide accountability for future utility-scale deployments, ensuring that any disturbance, whether incidental or minimal, is offset.

Conclusion

While I recognize that technology like this can play a role in California's water resilience efforts, environmental protection and monitoring cannot be compromised, even for a pilot project. By ensuring rigorous environmental safeguards in the early stages, the Commission can evaluate water solutions that strengthen community resilience and protect marine ecosystems. I support the deployment of the Iceberg buoy pilot. However, I strongly urge that the Commission require independent, third-party environmental monitoring and establish a hard-substrate mitigation fund to address benthic habitat impacts. These conditions will ensure that the project is implemented responsibly and provide critical data to inform future expansion.

Thank you for considering these comments. I appreciate the Commission's careful review to ensure that projects like the Iceberg buoy are executed with strong environmental oversight.

Sincerely,

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Acronyms

CCC - California Coastal Commission

CEQA - California Environmental Quality Act

EIR - Environmental Impact Report

IS/ MND - Initial Study/Mitigated Negative Declaration

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